

Internet Appendix

“Beyond All-Equity: Levered Diversification over the Lifecycle”

A Data appendix

This appendix describes the construction of the asset class returns. Section A.1 describes the country-year panel and the conversion of returns into the investor’s currency. Section A.2 describes gold returns. Section A.3 describes the construction of the managed-futures series and compares it with commercial trend-following indexes and funds.

A.1 Country-year panel and local real returns

The panel begins with release 6 of the Jordà-Schularick-Taylor Macrohistory Database (Jordà et al., 2023; Jordà et al., 2019). We extend each series through 2025 using dividend-adjusted local indexes, locally listed exchange-traded funds, and the Global Macro Database (Müller, Xu, Lehib, and Chen, 2025). A source indicator identifies the appended years.

Let $S_{c,f,t}$ denote units of the investor’s currency c per unit of foreign currency f . If $R_{f,t}^N$ is a nominal return in foreign currency and $\pi_{c,t}$ is local inflation, the local real return on an unhedged foreign asset is

$$1 + R_{f,t}^{(c)} = (1 + R_{f,t}^N) \frac{S_{c,f,t}}{S_{c,f,t-1}} \frac{1}{1 + \pi_{c,t}}. \quad (\text{A.1})$$

We apply this conversion to international stocks and gold. The international stock basket excludes the investor’s country and weights foreign markets by lagged market capitalization, constructed from the market-capitalization-to-GDP ratios of Kuvshinov and Zimmermann (2022), with real GDP weights from the Maddison Project (via the Macrohistory Database and Müller, Xu, Lehib, and Chen 2025) when these ratios are unavailable.

Figure A.1 plots the cumulative real return on each asset class for a US investor. The logarithmic scale makes proportional changes comparable across dates.

A.2 Gold

Gold returns use the annual US dollar price from MeasuringWorth (Officer and Williamson, 2024) through 1967 and the London Bullion Market Association afternoon fixing (London Bullion Market Association, 2025) from 1968. Both are quoted in dollars per troy ounce,

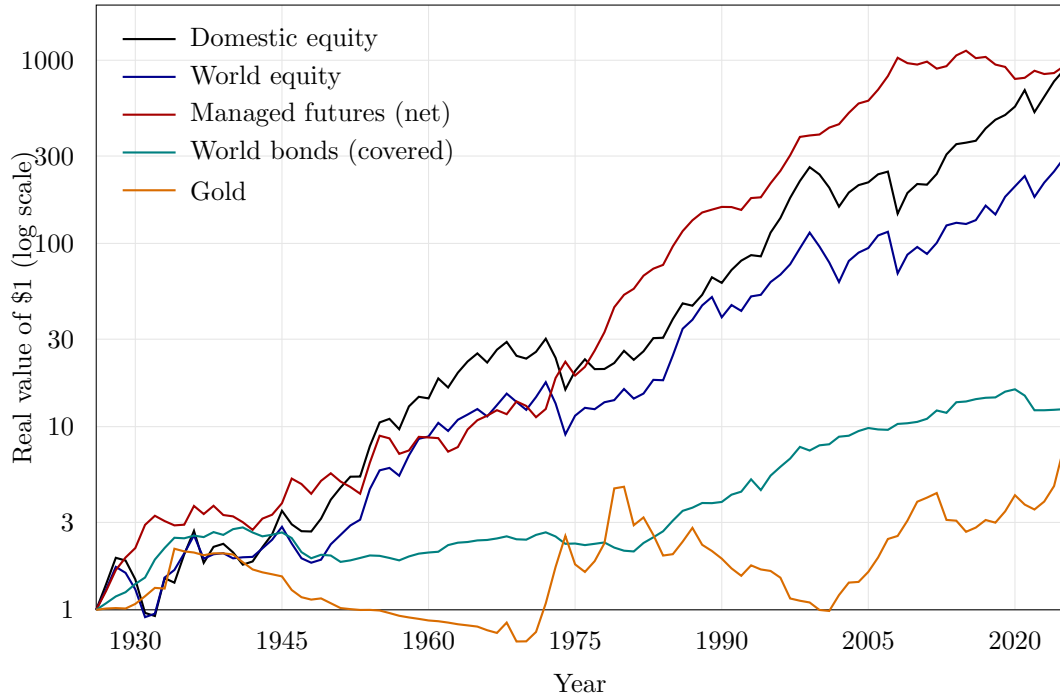


Figure A.1. Cumulative real returns by asset class. The figure shows the cumulative real return on each asset class for a US investor from 1927 to 2025 on a logarithmic scale. Managed futures are net of the 0.85% management fee and 0.61% trading costs. Global bonds are the hedged, equally weighted basket of all panel government bonds. Gold is net of custody costs. Stocks and gold are unhedged, and bonds and managed futures are hedged into US dollars.

and we splice them without adjustment. We deduct custody costs of 0.40% per year and convert the dollar return into local currency using equation (A.1). Before 1968, the dollar price was administered, and exchange-rate parities under the gold standard and the Bretton Woods system also constrained gold prices in other currencies. US residents regained the right to hold gold at the end of 1974 (United States Mint, 1974).

A.3 Managed futures

A.3.1 Monthly inputs

The managed-futures series is constructed from monthly data and then compounded into calendar years for the annual simulations. Table A.I describes the monthly inputs. The source families are OECD and FRED national equity and yield series (Organisation for Economic Co-operation and Development, 2026; Federal Reserve Bank of St. Louis, 2026), NBER historical series (National Bureau of Economic Research, 2016), World Bank commodity prices (World Bank, 2025), and Federal Reserve H.10 exchange rates (Board of Governors of the Federal Reserve System, 2026). Bond futures are approximated by synthetic excess returns on par ten-year bonds, equity futures by price index returns, and currency forwards by lagged three-month interest differentials. The first month of each source segment receives no return, so that differences in price levels across sources do not enter the trading signal. The series earns the US dollar cash return on its collateral once. For the annual simulations, its dollar net asset value is treated as a foreign asset and hedged in the same manner as global bonds.

A.3.2 Comparison with commercial trend-following indexes

Figure A.2 and Table A.II compare the monthly dollar returns of the managed-futures series with those of the SG CTA, SG Trend, and BTOP50 indexes of commercial funds and with the KMLMSIM simulation of the KFA MLM Index (Société Générale, 2026; BarclayHedge, 2026; Testfolio, 2026). These comparisons do not depend on the annual panel, the currency hedge, or the lifecycle simulations. The proprietary indexes are used only for this comparison. The managed-futures series has correlations of 0.55 to 0.59 with the three commercial indexes over 2000 to 2025, whereas the three indexes have correlations of 0.96 to 0.97 with each other.

Table A.I: Monthly data used to construct the managed-futures series

The table describes the monthly data sources, transformations, and markets used to construct the managed-futures series. Each source segment is documented in the replication files with its provider, series identifier, dates, and transformation.

Input	Monthly source and transformation	Markets
Equities	National price indexes from OECD Main Economic Indicators via FRED and, before their availability, the US, UK, German, and French NBER Macrobhistory monthly price series.	18 local-currency price returns; dividends are absent.
Rates	Monthly government bond yields from OECD via FRED, augmented before their start by NBER US yields, UK Consol and Bank of England yields, and the French 3% rente. Each yield is priced as a par ten-year bond, less the lagged three-month cash return.	18 synthetic local-currency bond excess returns.
Commodities	NBER and BLS monthly wholesale prices for the early history and the World Bank Pink Sheet from 1960. Gold uses the official US price before 1960 and the Pink Sheet thereafter; silver starts in 1960.	19 commodity spot returns plus gold and silver; carry and roll returns are unavailable.
Currencies	Federal Reserve H.10 dollar exchange rates via FRED at month-end, with a lagged three-month interest differential forming a synthetic forward return.	AUD, CAD, CHF, GBP, and JPY from 1971 and EUR from 1999.
Collateral and deflator	Monthly three-month cash rates from OECD via FRED, with the prior-year annual rate where monthly rates do not exist, and monthly US CPI.	US cash collateral return; CPI converts the final return to real terms.

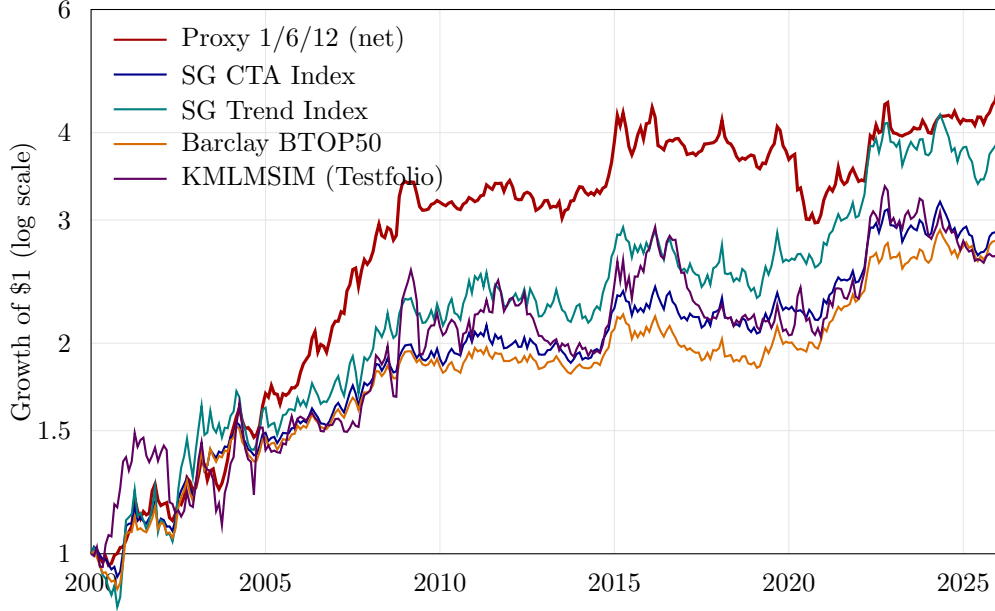


Figure A.2. Growth of one dollar in managed-futures strategies. The figure shows the cumulative monthly return from 2000 to 2025 on the managed-futures series, three commercial trend-following indexes, and KMLMSIM, the Testfolio simulation of the KFA MLM Index. The commercial indexes are net of fees, and the managed-futures series is net of its management fee and trading costs. KMLMSIM uses the KFA MLM Index less 0.90% per year before the inception of the KMLM fund.

Table A.II: Correlations of the managed-futures series with trend-following benchmarks
The table reports the correlation of monthly returns on the managed-futures series with the returns on commercial trend-following indexes and funds over the common period shown. KMLMSIM is the Testfolio simulation of the KFA MLM Index before the inception of the KMLM fund, and the rows for DBMF and KMLM use the funds' own returns from the first full month after listing (Testfolio, 2026; Yahoo Finance, 2026).

Benchmark	Corr.	Common window
SG CTA Index	0.57	2000-01–2025-12 (312 mo.)
SG Trend Index	0.55	2000-01–2025-12 (312 mo.)
Barclay BTOP50 Index	0.59	2000-01–2025-12 (312 mo.)
KMLMSIM (Testfolio simulation)	0.53	2000-01–2025-12 (312 mo.)
DBMF ETF (live)	0.61	2019-06–2025-12 (79 mo.)
KMLM ETF (live)	0.63	2021-01–2025-12 (60 mo.)

Memo, 2000–2025: the three commercial indexes correlate 0.96–0.97 with each other, whereas KMLMSIM correlates 0.60 with SG CTA. The proxy correlates 0.55–0.59 with the three commercial indexes.

A.3.3 Comparison with investable funds

Tables A.III to A.V extend the comparison to investable managed-futures funds with at least 60 months of returns. Shorter histories provide imprecise estimates. The pairwise correlations in Table A.III use each pair’s overlapping months, and Table A.IV uses a common window for all series with at least ten years of returns. Table A.V regresses the returns of each index and fund on the returns of the managed-futures series. The regressions have R^2 values of about 0.3 for the commercial indexes and intercepts that are not statistically different from zero.

A.3.4 Annual properties

Table A.VI reports the annual properties of the managed-futures series and the other asset classes for a US investor, and Figure A.3 plots the annual return on managed futures against the annual return on each asset class. The 10% volatility target applies ex ante to monthly returns, using the preceding 36 months of returns (at least 24), and the scaling factor reaches its cap of three in 26.4% of months. The annualized standard deviation of monthly real gross returns is 10.71% over 1927 to 2025. Compounding into calendar years gives an annual standard deviation of 14.18% net of costs, because monthly returns are not independent.

A.3.5 Alternative construction choices

Table A.VII reports the properties of the managed-futures series under alternative signal horizons and management fees. Each variant is charged its own trading costs. Panel E of Table VII in the paper reports the lifecycle outcomes of each variant, rebuilt through the same currency hedge and evaluated with a common flat trading cost at the equal-volatility exposures.

Table A.III: Correlations of monthly returns: managed-futures series, benchmarks, and investable funds

The table reports the correlations of monthly returns between the managed-futures series (Proxy), commercial trend-following benchmarks, and investable funds with at least 60 months of returns. Each correlation uses the overlapping months of the two series within January 2000 to December 2025. The funds' first months are June 2019 for DBMF, January 2021 for KMLM, February 2011 for WTMF, August 2013 for QMHIX, and September 2014 for AHLIX. Fund returns are total returns net of fees from Yahoo Finance adjusted closing prices. QMHIX, AHLIX, DBMF, and KMLM are trend-following funds; WTMF is a systematic equity-timing strategy. Shading is proportional to the correlation.

	SG CTA	SG Trend	BTOP50	KMLMSIM	DBMF	KMLM	QMHIX	AHLIX	WTMF
Proxy	0.57	0.55	0.59	0.53	0.61	0.63	0.53	0.56	0.19
SG CTA	—	0.97	0.97	0.60	0.86	0.77	0.83	0.84	0.40
SG Trend	0.97	—	0.96	0.60	0.86	0.74	0.85	0.86	0.42
BTOP50	0.97	0.96	—	0.58	0.85	0.73	0.79	0.83	0.41
KMLMSIM	0.60	0.60	0.58	—	0.66	1.00	0.75	0.63	0.34
DBMF	0.86	0.86	0.85	0.66	—	0.68	0.73	0.74	0.24
KMLM	0.77	0.74	0.73	1.00	0.68	—	0.74	0.65	0.12
QMHIX	0.83	0.85	0.79	0.75	0.73	0.74	—	0.70	0.33
AHLIX	0.84	0.86	0.83	0.63	0.74	0.65	0.70	—	0.39
WTMF	0.40	0.42	0.41	0.34	0.24	0.12	0.33	0.39	—

Table A.IV: Correlations of monthly returns over a common window

The table reports the correlations of monthly returns between the managed-futures series and the benchmarks and funds with at least ten years of returns, all measured over the common window from September 2014 to December 2025 (136 months). Sources are as in Table A.III.

	SG CTA	SG Trend	BTOP50	KMLMSIM	WTMF	QMHIX	AHLIX
Proxy	0.58	0.59	0.62	0.54	0.20	0.53	0.56
SG CTA	—	0.98	0.97	0.72	0.39	0.84	0.84
SG Trend	0.98	—	0.96	0.72	0.40	0.86	0.86
BTOP50	0.97	0.96	—	0.69	0.41	0.79	0.83
KMLMSIM	0.72	0.72	0.69	—	0.31	0.75	0.63
WTMF	0.39	0.40	0.41	0.31	—	0.33	0.39
QMHIX	0.84	0.86	0.79	0.75	0.33	—	0.70
AHLIX	0.84	0.86	0.83	0.63	0.39	0.70	—

Table A.V: Regressions of benchmark and fund returns on the managed-futures series

The table reports ordinary least squares regressions $R_t^s = \alpha + \beta R_t^{\text{proxy}} + \varepsilon_t$ of monthly returns on each benchmark and fund on the returns of the managed-futures series, over each series' overlap with the managed-futures series within January 2000 to December 2025. α is in percent per month, and t -statistics are in parentheses. σ_f/σ_p is the ratio of annualized volatilities, and TE is the tracking error, the annualized standard deviation of $R_t^s - R_t^{\text{proxy}}$, in percent per year.

Series	n	α (%/mo.)	β	R^2	σ_f/σ_p	TE (%/yr.)
SG CTA	312	+0.12 (+1.0)	0.47 (+12.1)	0.32	0.83	9.1
SG Trend	312	+0.13 (+0.7)	0.70 (+11.5)	0.30	1.27	11.6
BTOP50	312	+0.12 (+1.1)	0.44 (+12.7)	0.34	0.76	8.7
KMLMSIM	312	-0.01 (-0.0)	0.79 (+11.1)	0.28	1.48	13.4
DBMF	79	+0.49 (+1.7)	0.63 (+6.8)	0.38	1.02	9.7
KMLM	60	-0.11 (-0.3)	0.82 (+6.1)	0.39	1.31	10.1
QMHIX	149	+0.25 (+0.8)	0.76 (+7.5)	0.28	1.45	13.2
AHLIX	136	+0.26 (+1.3)	0.49 (+7.9)	0.32	0.88	9.7
WTMF	179	+0.03 (+0.3)	0.12 (+2.6)	0.04	0.63	10.8

Table A.VI: Annual properties of the managed-futures series and other asset classes

The table summarizes annual real returns for a US investor from 1927 to 2025. The first panel reports the arithmetic mean, the standard deviation (population), the ratio of mean to standard deviation, skewness, the 5th percentile, and the maximum drawdown in cumulative real value. The second panel reports the returns on world stocks, managed futures, and gold and US inflation in the ten years with the lowest world stock returns. Unlike Table I, the table uses the 99 US observations rather than the pooled panel.

Asset	Mean	SD	Mean/SD	Skew	5th pctl	Max DD
Domestic equity	8.81%	18.69%	0.47	-0.24	-21.4%	-51.9%
World equity	7.42%	17.48%	0.42	-0.25	-23.1%	-48.1%
World bonds (covered)	2.82%	7.03%	0.40	+0.15	-7.6%	-35.2%
Gold	3.70%	19.70%	0.19	+1.88	-21.2%	-79.0%
Managed-futures proxy (net)	8.05%	14.18%	0.57	+0.52	-11.5%	-29.9%

Year	World equity	MF proxy	Gold	U.S. inflation
1930	-18.7%	12.6%	6.0%	-2.3%
1931	-29.5%	33.8%	10.2%	-9.0%
1937	-25.4%	-9.6%	-3.4%	3.6%
1946	-18.9%	36.2%	-15.7%	8.3%
1973	-23.5%	47.6%	58.1%	6.3%
1974	-32.2%	22.6%	47.6%	11.0%
1990	-22.5%	3.5%	-9.2%	4.9%
2002	-21.8%	4.4%	22.0%	1.6%
2008	-40.7%	25.8%	3.9%	3.8%
2022	-23.1%	9.2%	-6.0%	8.0%

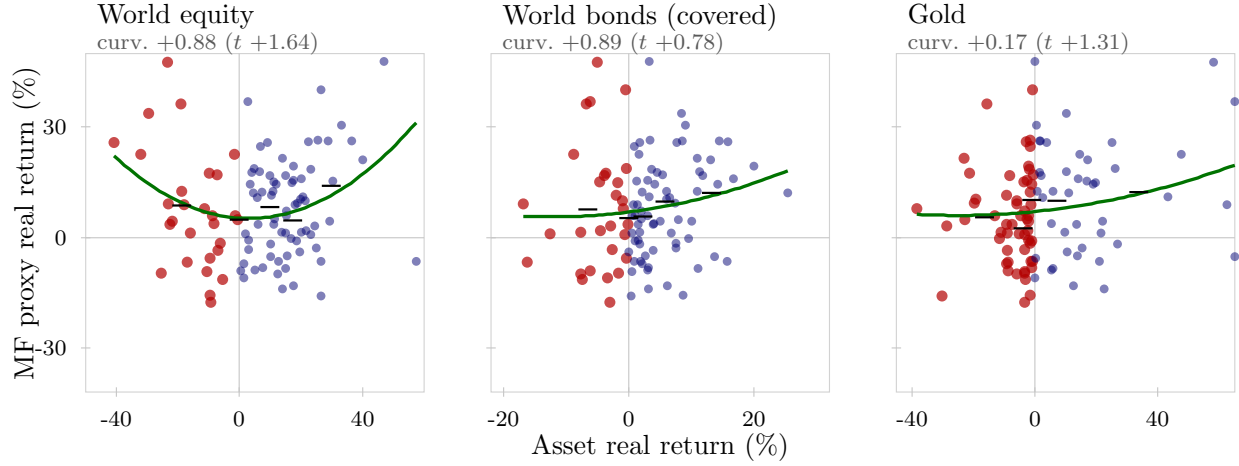


Figure A.3. Annual managed-futures returns and asset class returns. The figure plots the annual real return on managed futures against the annual real return on each asset class for a US investor from 1927 to 2025. Red (blue) points are years in which the asset class lost (gained). The green curve is a quadratic fit, and the black marks are average managed-futures returns within quintiles of the asset class return. “curv.” is the coefficient on the squared term, and t is its heteroskedasticity-robust t -statistic.

Table A.VII: Managed-futures series under alternative construction choices

The table reports annual real returns for a US investor from 1927 to 2025 on variants of the managed-futures series that differ in signal horizons and management fees, net of each variant’s fee and realized trading costs. The baseline mean differs slightly from Table A.VI, which uses a flat annual trading-cost deduction. Mean/SD uses the sample standard deviation. “Corr. baseline” is the correlation of each variant’s annual returns with the 1/6/12-month baseline, marked † .

Variant	Mean	SD	Mean/SD	Skew	5th pctl	Corr. baseline
1/6/12 blend (baseline) †	7.91%	13.98%	0.57	+0.45	-11.8%	1.000
1/3/12 blend	7.94%	14.21%	0.56	+0.66	-11.2%	0.957
12-month signal only	8.19%	14.58%	0.56	+0.21	-14.3%	0.831
1-month signal only	5.57%	12.53%	0.44	+0.61	-10.6%	0.759
1/6/12 blend, fee doubled to 1.70%	6.98%	13.86%	0.50	+0.45	-12.6%	1.000

B Methods appendix

This appendix provides additional details on the portfolio returns, the bootstrap, the risk parity weights, the properties of the household problem, and the experiments reported in the paper.

B.1 Currency hedge and portfolio returns

Let A_t^f be the gross nominal return on a foreign asset, B_t^f and B_t^c the gross nominal bill returns in the foreign and local currencies, and $q_t = S_{c,f,t}/S_{c,f,t-1}$ the change in the exchange rate. The hedge sells the initial foreign principal forward at the covered-interest-parity rate, so the local real return on the hedged position is

$$1 + R_t^{c,H} = \frac{B_t^c/B_t^f + (A_t^f - 1)q_t}{1 + \pi_{c,t}}. \quad (\text{B.1})$$

Gains and losses beyond the initial principal remain exposed to the exchange rate. Realized annual bill returns proxy for the interest rates that price the forward. The managed-futures series already includes its US dollar collateral return, and the forward payoff is added according to equation (B.1). The hedging cost of 0.10% per year applies to the hedged notional.

For a levered portfolio, the coefficient on the local bill return in equation (2) of the paper is negative. Holding asset returns fixed, a higher ex post real bill return lowers the portfolio return, and an inflation surprise that lowers the real bill return raises it.

B.2 Bootstrap continuation and local currencies

When a bootstrap block reaches a gap or the end of a country's data, the block continues at the first available year of another country, selected through a uniformly drawn panel row. Countries are therefore weighted by the length of their samples. Each country-year contains returns measured in that country's currency and price index, so a block from Japan in the 1980s is a consistent sequence of yen-denominated real returns. A simulated lifetime can join real return histories from several countries. The simulation does not model a currency exchange at these joins.

B.3 Risk parity weights

For a positive definite covariance matrix $\widehat{\Sigma}$, the risk parity weights solve

$$x^* = \arg \min_{x_i > 0} \left\{ \frac{1}{2} x' \widehat{\Sigma} x - \frac{1}{4} \sum_{i=1}^4 \log x_i \right\}, \quad v = x^* / (\mathbf{1}' x^*). \quad (\text{B.2})$$

The first-order conditions imply $x_i(\widehat{\Sigma}x)_i = 1/4$, so each asset class contributes equally to portfolio variance (Maillard, Roncalli, and Teiletche, 2010; Roncalli, 2014). The covariance matrix is the sample covariance of annual real returns on the four asset classes, net of fees, custody, and hedging costs, across all 1,561 country-years with equal weights. We apply no shrinkage or winsorization. All eigenvalues are positive, and the computed risk contributions differ from 25% by less than 10^{-7} . The weights are 14.53% stocks, 47.47% global bonds, 12.05% gold, and 25.95% managed futures. Borrowing costs are added only after the weights are scaled to the target exposure, so the equal contributions apply to the unlevered portfolio.

For the comparison at equal volatility, we scale the weights of each diversified strategy by the gross exposure G at which the sample standard deviation of its annual real return, including financing and hedging costs, equals the 25.28% of the unlevered all-equity strategy across the 1,561 country-years. We solve for G by bisection. The resulting exposures are 166% for the proportional strategy, 265% for risk parity, 126% for stocks, gold, and managed futures, and 132% for stocks and managed futures. When we reduce managed-futures returns, we hold these exposures fixed.

B.4 Properties of the household problem

This section derives the two properties of the household problem used in the paper. Let v denote the asset class proportions, G gross exposure, and r_c the savings rate, and let $V(r_c, v, G) = E[U(C, B)]$ denote expected utility.

Invariance of ruin to the savings rate. For a fixed strategy and simulated path, wealth is proportional to the savings rate. Let $A(v, G)$ be retirement wealth, $d_t(v, G)$ the withdrawal actually paid in retirement, and $b(v, G)$ the bequest, all at a unit savings rate. Then

$$D = 0.04 r_c A, \quad C_t = \max\{SS_t + r_c d_t, SS I_t\}, \quad B = r_c b. \quad (\text{B.3})$$

The planned and actual withdrawals scale together, so the event of exhausting wealth does not depend on $r_c > 0$. Table C.VII shows that ruin probabilities are identical across savings

rates. Fixing the withdrawal at 4% of the all-equity strategy's retirement wealth breaks this proportionality, which is why the path-matched income experiment measures a distinct outcome.

Equivalent savings rates and marginal utility of return. V increases with r_c whenever additional saving raises consumption above the floor or the bequest. At a locally unique equivalent savings rate with $V_{r_c} > 0$, the implicit function theorem gives, for an allocation parameter a ,

$$\frac{\partial r_c^*}{\partial a} = -\frac{V_a}{V_{r_c}}. \quad (\text{B.4})$$

Let $\tilde{R}_t$ be the vector of net asset class returns and $M_t = \partial U / \partial R_t^p$ the sensitivity of lifetime utility to the portfolio return in year t , which includes all subsequent effects on wealth, withdrawals, consumption, and the bequest. For a change in proportions h with $\mathbf{1}'h = 0$, and under conditions that allow differentiation inside the expectation (Glasserman, 1991),

$$V_h = G \sum_t \left(E[M_t] E[h' \tilde{R}_t] + \text{Cov}(M_t, h' \tilde{R}_t) \right), \quad (\text{B.5})$$

$$V_G = E \left[\sum_t M_t (v' \tilde{R}_t - R_t^f - \phi) \right], \quad G > 1. \quad (\text{B.6})$$

Equation (B.5) separates the value of a change in weights into an average-return term and a term that measures whether the added asset pays off when the marginal utility of return is high. Equation (B.6) shows that leverage adds value only when the utility-weighted portfolio return exceeds the bill return plus the borrowing spread. Differentiability fails on paths where the household's wealth is exhausted or the consumption floor begins to bind, and such paths occur with positive probability in a discrete bootstrap. The derivatives therefore describe small changes in weights, and the paper reports exact changes. The risk parity conditions do not imply that either derivative is zero, so the risk parity weights need not maximize household utility even with a known covariance matrix.

B.5 Financing routes

The borrowing spread of 0.30% is an annual spread above the local bill rate. Futures and box spreads on European-style index options provide financing close to this level (CME Group, 2024). Wang (2025) reports implied box-spread financing rates of 0.30% to 0.50% above three- and five-year Treasury yields in February 2025. Access, maturity, collateral requirements, and execution costs differ across these routes. The model separately charges

managed-futures fees and trading costs, gold custody, and hedging costs, and it does not charge an additional cost for annual rebalancing. The volatility-scaled positions inside the managed-futures series are distinct from the portfolio’s gross exposure, which measures the allocation to each asset class.

Holding leveraged fund shares differs from borrowing against pledged assets. Direct borrowing can force liquidation or require additional collateral, whereas a leveraged fund can reduce exposure internally. The annual model sets depleted wealth to zero and does not simulate forced liquidation within the year.

B.6 Resampled calendar histories

The analysis in Table VI measures uncertainty about the historical record rather than simulation error given that record. We resample blocks of calendar-year cross-sections from the panel. Each cross-section retains all countries observed in that year and their joint returns. The outer blocks follow the stationary block bootstrap of Politis and Romano (1994) with circular wrapping and average lengths of 5, 10, or 20 years. Each of 100 outer histories contains 99 calendar years, which are relabeled consecutively before the household simulation. Each history supports 1,000 household simulations with the ten-year average block length of the paper. The strategies are held at their base-case equal-volatility exposures in every history.

B.7 Historically investable asset classes

The historical availability analysis retains the 1927 to 2025 sample but sets the gold weight to zero before 1975 and adds managed-futures sectors as their futures markets opened, starting from commodities and adding currencies in 1972, interest rates in 1977, and equity indexes in 1982 (CME Group, 2026; United States Mint, 1974). The removed weight is reallocated in proportion to the remaining weights at unchanged gross exposure. For 1960 to 1963, the restricted managed-futures series lacks twelve months of signals, and its weight is also reallocated. At 175% exposure, the restricted proportional strategy has a ruin probability of 2.26% and an equivalent savings rate of 5.77%.

C Supplemental results

This appendix contains supplemental empirical results.

C.1 Removing one asset class

In Section 5.3, we examine the contribution of each asset class. Table C.I reports retirement outcomes after removing one asset class from the proportional and risk parity strategies at 175% exposure, with the removed weight redistributed across the remaining asset classes. Removing managed futures raises the ruin probability, the equivalent savings rate, and the path-matched income failure probability of both strategies. Removing bonds improves all three outcomes. Removing gold lowers the equivalent savings rate and the path-matched income failure of both strategies, with little change in the ruin probability.

C.2 First-order decomposition

Tables C.II to C.IV evaluate the derivatives in equations (B.5) and (B.6) at the initial weights and compare them with exact changes. The derivatives are informative only for small reallocations. For reallocations of 100% of wealth or more, such as removing all three diversifying asset classes, the local and exact values can differ in sign. Where the two agree in sign, adding any diversifying asset class to the levered all-equity strategy lowers the equivalent savings rate through the covariance term. The average-return term raises it for bonds and gold and is close to zero for managed futures. For leverage, the average-return term lowers the equivalent savings rate and the covariance term offsets part of this effect. Beyond 175% exposure, the covariance term dominates for the all-equity strategy. Table C.III reports the exact changes from leverage in both samples.

C.3 Risk parity weights from alternative samples

Table C.V reports the risk parity weights estimated from the post-1970 sample and from the sample that excludes Italy in 1942. The lifecycle simulations in Table VII retain the full-sample weights, so that they measure the effect of the evaluation sample rather than of the estimation sample.

C.4 Risk aversion

Table C.VI reports equivalent savings rates for the all-equity and proportional strategies at 200% exposure for risk aversion parameters from 2 to 10. We consider two conventions for

the bequest intensity. The joint calibration holds fixed the monthly intensity of Anarkulova, Cederburg, and O’Doherty (2025), so that the annual intensity changes with γ . The fixed calibration holds the annual intensity at $\theta = 2,360$. The proportional strategy requires less saving than the unlevered all-equity strategy under every specification.

The equivalent savings rates of the levered all-equity strategy vary non-monotonically with γ for two reasons. First, at $\gamma = 3$ the joint calibration implies a bequest intensity about eight times larger than the fixed calibration, and the bequest accounts for 52% of the negative expected utility of the levered all-equity strategy. The frequent ruin of this strategy leaves no bequest, which explains its equivalent savings rate of 20.09% under the joint calibration, compared with 12.42% under the fixed calibration. Second, as γ rises, expected utility is increasingly determined by the lowest consumption paths. The worst 1% of lifetimes account for 19% of the negative expected utility of the levered all-equity strategy at $\gamma = 3.84$, 30% at $\gamma = 5$, 48% at $\gamma = 7.5$, and 61% at $\gamma = 10$. The equivalent savings rate then reflects a small number of paths in which consumption falls toward the Social Security and SSI floor, and it need not be monotonic in γ . For $\gamma \geq 5$, the bequest accounts for less than 0.02% of expected utility, and the two conventions give the same equivalent savings rates.

C.5 Savings and withdrawal rates

Table C.VII varies the savings rate from 5% to 15% and the withdrawal rate from 3% to 5%. The ruin probabilities do not change with the savings rate, as implied by equation (B.3). Including the all-equity strategy at 200% exposure distinguishes the effect of diversification from that of leverage.

C.6 Real bill rates and the returns to leverage

Table C.VIII decomposes the annual return gain from raising exposure from 100% to 175% by quintile of the ex post real bill return. The bill leg contributes positively in years with negative real bill returns, but the total gain is positive in all five quintiles for both the all-equity and proportional strategies.

C.7 Margin requirements

The 25% threshold corresponds to the maintenance requirement for long margin-eligible equities in US Regulation T accounts (Financial Industry Regulatory Authority, 2021). The

equity fraction of an account with gross assets L and debt $L - 1$ at the start of the year is

$$m_{\text{end}} = \frac{LG_A - (L - 1)G_D}{LG_A} \quad (\text{C.1})$$

at year-end, where G_A is the gross nominal asset return and G_D is the gross nominal cost of debt, including the borrowing spread. Table C.IX reports year-end breaches in the bootstrap simulations. Table C.X uses monthly US dollar data from 1970 to 2025 to compare month-end and year-end tests on the same paths.

C.8 Personal taxes

Our returns are net of implementation costs but before personal taxes. Gold, managed-futures distributions, rebalancing gains, and interest expense can be taxed differently. For example, the tax disclosure of the SPDR Gold Trust treats individual holders as owning their share of the underlying gold, with long-term gains taxed as collectibles (SPDR Gold Trust, 2022; Internal Revenue Service, 2026), and managed-futures fund prospectuses report returns after taxes on distributions (Litman Gregory Funds Trust, 2026). Rankings in taxable accounts can therefore differ from those in tax-advantaged retirement accounts.

Table C.I: Removing one asset class

The table reports retirement outcomes for the proportional (Panel A) and risk parity (Panel B) strategies at 175% exposure after removing one asset class. The weight of the removed asset class is redistributed across the remaining asset classes in proportion to their weights, holding gross exposure fixed. Each comparison is based on 10,000 bootstrap simulations of household outcomes that share return, earnings, and mortality draws. The household saves 10% of labor income. Path-matched income failure is the probability of failing to fund 4% of the retirement wealth of the all-equity strategy at 100% exposure. The sample is 1927 to 2025.

Asset class removed	Ruin probability	Equivalent savings rate	Path-matched income failure
Panel A: Proportional strategy at 175% exposure			
None	2.13%	5.87%	2.17%
Global bonds	1.92%	4.15%	0.86%
Gold	2.12%	5.27%	0.99%
Managed futures	3.71%	7.79%	6.28%
Panel B: Risk parity strategy at 175% exposure			
None	3.79%	8.29%	14.25%
Global bonds	1.88%	3.96%	1.78%
Gold	3.93%	8.05%	11.70%
Managed futures	8.09%	14.03%	36.68%

Table C.II: First-order decomposition of changes in asset class weights

The table decomposes the changes in Panels A to C of Table V using the first-order approximation in Section B.4. Local is the derivative of the equivalent savings rate with respect to the change in weights, evaluated at the initial weights. The first three columns scale it to a reallocation of 10% of wealth and split it into a mean term, $E[M_t] E[\Delta R_t]$, and a covariance term, $\text{Cov}(M_t, \Delta R_t)$. The next two columns report the local value for the full reallocation and the exact change from Table V. Sign indicates whether the two agree in sign. The sample is 1927 to 2025, and entries are in percentage points.

Change in weights	Realloc.	Local per 10 pp	of which mean	of which cov.	Local, full move	Exact	Sign
Panel A: Add one asset class to the all-equity strategy at 175%							
+ Global bonds	46.7	-1.6	+2.4	-4.0	-7.6	-2.1	✓
+ Gold	29.2	-3.4	+1.6	-5.0	-9.9	-3.3	✓
+ Managed futures	29.2	-4.8	+0.2	-5.0	-14.0	-5.4	✓
+ All three asset classes	105.0	-3.0	+1.5	-4.5	-31.5	-3.8	✓
Panel B: Remove one asset class from the proportional strategy at 175%							
- Global bonds	46.7	-1.0	-1.6	+0.6	-4.5	-1.7	✓
- Gold	29.2	-0.9	-1.1	+0.2	-2.5	-0.8	✓
- Managed futures	29.2	+0.3	-0.1	+0.5	+0.9	+1.2	✓
- Bonds and gold	75.8	-0.9	-1.4	+0.5	-7.1	-1.6	✓
- All three asset classes	105.0	-0.6	-1.0	+0.5	-6.1	+3.8	×
Panel C: Remove one asset class from the risk parity strategy at 175%							
- Global bonds	83.1	-1.5	-1.7	+0.1	-12.6	-4.7	✓
- Gold	21.1	-1.2	-1.1	-0.1	-2.5	-1.4	✓
- Managed futures	45.4	-0.2	-0.1	-0.1	-1.1	+0.9	×
- Bonds and gold	104.2	-1.4	-1.5	+0.1	-15.1	-4.9	✓
- All three asset classes	149.6	-1.1	-1.1	+0.0	-16.1	+1.4	×

Table C.III: Changes in the equivalent savings rate from additional leverage

The table reports the change in the equivalent savings rate, in percentage points, from raising the gross exposure of each strategy. A negative change means that the household needs to save less. Each comparison is based on 10,000 bootstrap simulations. The last column draws returns from 1970 to 2025, after the 1968 separation of the free London gold market from the official price.

Gross exposure	Full sample (1927–2025)	Post-1970 sample (1970–2025)
Panel A: All-equity strategy		
100→125	−1.4	−0.9
100→175	−0.4	+2.5
175→200	+6.0	+17.3
Panel B: Proportional strategy		
100→125	−2.9	−2.3
100→175	−6.5	−5.3
175→200	−1.1	−0.9
Panel C: Risk parity strategy		
100→125	−3.5	−2.6
100→175	−8.0	−6.0
175→200	−1.5	−1.2

Table C.IV: First-order decomposition of changes in gross exposure

The table decomposes the changes in Table C.III using the first-order approximation in Section B.4. Local is the derivative of the equivalent savings rate with respect to gross exposure, evaluated at the initial exposure and scaled to an increase of 10% of wealth. The mean column is the average excess-return term, and the covariance column is the utility cost of the added volatility. The last three columns report the local value for the full step, the exact change, and whether the two agree in sign. The sample is 1927 to 2025, and entries are in percentage points.

Gross exposure	Local per 10 pp	of which mean	of which cov.	Local, full step	Exact	Sign
Panel A: All-equity strategy						
100→125	−0.8	−2.2	+1.4	−2.1	−1.4	✓
100→175	−0.8	−2.2	+1.4	−6.2	−0.4	✓
175→200	+1.3	−2.9	+4.2	+3.2	+6.0	✓
Panel B: Proportional strategy						
100→125	−1.1	−1.4	+0.3	−2.8	−2.9	✓
100→175	−1.1	−1.4	+0.3	−8.3	−6.5	✓
175→200	−0.9	−1.4	+0.5	−2.2	−1.1	✓
Panel C: Risk parity strategy						
100→125	−1.0	−1.1	+0.1	−2.4	−3.5	✓
100→175	−1.0	−1.1	+0.1	−7.2	−8.0	✓
175→200	−0.8	−1.1	+0.3	−2.1	−1.5	✓

Table C.V: Risk parity weights estimated from alternative samples

The table reports the unlevered risk parity weights that equalize the contributions of the four asset classes to portfolio variance, estimated from the sample covariance matrix of annual real returns in each sample. The weights sum to 100% before exposure scaling. All lifecycle simulations use the full-sample weights.

Estimation sample	Stocks	Global bonds	Gold	Managed futures
Full sample, 1927–2025	14.53%	47.47%	12.05%	25.95%
Post-1970 sample	19.62%	40.46%	15.38%	24.53%
Excluding Italy 1942	19.72%	42.22%	17.07%	20.99%

Table C.VI: Equivalent savings rates under alternative risk aversion

The table reports equivalent savings rates for the all-equity and proportional strategies at 200% exposure under alternative coefficients of relative risk aversion. The reference is the all-equity strategy at 100% exposure and a 10% savings rate under the same preferences. The joint calibration sets the annual bequest intensity to $\theta(\gamma) = 2,360 \times 12^{3.84-\gamma}$, which holds fixed the monthly intensity used by Anarkulova, Cederburg, and O’Doherty (2025); the fixed calibration holds the annual intensity at $\theta = 2,360$. Ruin probabilities do not depend on preferences and equal 6.98%, 20.28%, and 1.80% for the all-equity strategy at 100% and 200% and the proportional strategy at 200%. Each entry is based on 10,000 bootstrap simulations and solves for the equivalent savings rate by bisection on [0%, 100%]; no solution lies at a bound.

Bequest intensity	γ	All-equity 200%	Proportional 200%
Joint calibration	2	10.35	2.04
Joint calibration	3	20.09	3.38
Joint calibration	3.84	15.66	4.72
Joint calibration	5	20.25	4.54
Joint calibration	7.5	20.26	4.30
Joint calibration	10	17.60	4.21
Fixed annual bequest	2	9.23	2.60
Fixed annual bequest	3	12.42	4.60
Fixed annual bequest	3.84	15.66	4.72
Fixed annual bequest	5	20.26	4.54
Fixed annual bequest	7.5	20.26	4.30
Fixed annual bequest	10	17.60	4.21

Table C.VII: Equivalent savings rates under alternative contribution and withdrawal rates
The table reports ruin probabilities and equivalent savings rates for the all-equity and proportional strategies at 200% exposure under alternative savings and withdrawal rates. The first three rows vary the savings rate with a 4% withdrawal rate; the last three rows vary the withdrawal rate with a 10% savings rate. The base case appears in both groups. In each row, the reference is the all-equity strategy at 100% exposure with the savings and withdrawal rates of that row. Each row is based on 10,000 bootstrap simulations.

Savings rate	Withdrawal rate	All-equity 100% ruin	All-equity 200% ruin	All-equity 200% equiv.	Proportional 200% ruin	Proportional 200% equiv.
5.00	4.00	6.98	20.28	4.63	1.80	2.47
10.00	4.00	6.98	20.28	15.66	1.80	4.72
15.00	4.00	6.98	20.28	46.06	1.80	6.84
10.00	3.00	2.54	13.73	9.84	0.92	5.09
10.00	4.00	6.98	20.28	15.66	1.80	4.72
10.00	5.00	15.47	26.75	24.11	4.37	4.00

Table C.VIII: Incremental returns from leverage by real bill rate quintile

The table decomposes the difference in annual real return between each strategy at 175% and at 100% exposure into contributions from the asset classes, the bill leg of borrowing, the borrowing spread, and the currency-hedge friction. Country-years are sorted into quintiles by their ex post real bill return, with Q1 containing the lowest real bill returns. For weights a_i summing to one, the four components are $0.75 \sum_i a_i R_i$, $-0.75 R_f$, -0.75ϕ , and $-0.75 h \kappa$, with $\phi = 0.30\%$, $\kappa = 0.10\%$, and h the hedged share at 100% exposure. The sample is the 1,561 country-years from 1927 to 2025, and entries are in percentage points.

Quintile	n	Mean bill	Asset classes	Bill leg	Spread	Hedge	Total
<i>All-equity: 175% minus 100% exposure</i>							
Q1	313	-7.47	2.85	5.60	-0.23	0.00	8.22
Q2	312	-0.84	5.31	0.63	-0.23	0.00	5.72
Q3	312	0.77	6.82	-0.58	-0.23	0.00	6.02
Q4	312	2.41	7.90	-1.80	-0.23	0.00	5.87
Q5	312	6.44	7.56	-4.83	-0.23	0.00	2.50
<i>Proportional: 175% minus 100% exposure</i>							
Q1	313	-7.47	0.33	5.60	-0.23	-0.03	5.67
Q2	312	-0.84	3.62	0.63	-0.23	-0.03	3.99
Q3	312	0.77	4.87	-0.58	-0.23	-0.03	4.04
Q4	312	2.41	5.84	-1.80	-0.23	-0.03	3.77
Q5	312	6.44	6.73	-4.83	-0.23	-0.03	1.64

Table C.IX: Year-end maintenance margin breaches

The table reports the number of year-end breaches of a 25% maintenance margin requirement across 10,000 simulated 86-year account paths (860,000 account-years) with annual rebalancing to the target exposure. Asset and debt values are nominal. The simulation excludes contributions, withdrawals, and mortality, and the account returns to its target exposure after each breach. The last column reports the lowest year-end ratio of equity to assets.

Strategy	Gross exposure	Breaches	Minimum equity share
All-equity	175.00%	1,081	21.41%
All-equity	200.00%	17,934	8.31%
Proportional	200.00%	0	38.59%

Table C.X: Month-end and year-end maintenance margin breaches

The table reports the number of calendar years with at least one breach of a 25% or 40% maintenance margin requirement for the all-equity and proportional strategies, measured at month-end (Monthly years) or at year-end (Annual years) on the same asset and debt paths. The sample covers the 56 complete years from 1970 to 2025 in US dollars. Stocks are 33/67 US and non-US total-return series, global bonds are proxied by intermediate US Treasuries, gold is the spot price less custody, and managed futures are the net proxy. Debt accrues the US bill rate plus 0.30% per year. Asset quantities are fixed within each year, and exposure returns to target at each year-end. The last column reports the lowest equity-to-asset ratio.

Portfolio	Gross (%)	Maintenance	Monthly years	Annual years	Min. equity share
All-equity	100	25.00%	0	0	100.00%
All-equity	100	40.00%	0	0	100.00%
All-equity	125	25.00%	0	0	62.51%
All-equity	125	40.00%	0	0	62.51%
All-equity	150	25.00%	0	0	37.52%
All-equity	150	40.00%	1	0	37.52%
All-equity	175	25.00%	1	1	19.67%
All-equity	175	40.00%	3	2	19.67%
All-equity	200	25.00%	1	1	6.29%
All-equity	200	40.00%	13	9	6.29%
Proportional	100	25.00%	0	0	100.00%
Proportional	100	40.00%	0	0	100.00%
Proportional	125	25.00%	0	0	76.13%
Proportional	125	40.00%	0	0	76.13%
Proportional	150	25.00%	0	0	60.21%
Proportional	150	40.00%	0	0	60.21%
Proportional	175	25.00%	0	0	48.85%
Proportional	175	40.00%	0	0	48.85%
Proportional	200	25.00%	0	0	40.32%
Proportional	200	40.00%	0	0	40.32%

D Replication

The [replication repository](#) contains the public source data, strategy definitions, numerical results, and code to rebuild all exhibits. The [annual return panel](#) and the [monthly managed-futures data](#) are the two data inputs. The proprietary SG and BarclayHedge indexes are used only in Section [A.3.2](#), and their redistribution is subject to the providers' terms.

The [experimental protocol](#) specifies the risk parity estimation, the exposure grid, and the sensitivity analyses. The [result files](#) contain the risk parity weights, covariance matrix, input checksums, and the complete results for each specification. The simulations use 10,000 paths, seed 20260827, and an average block length of ten years. Each alternative sample recomputes the utility reference of the unlevered all-equity strategy. The [validation checks](#) confirm, among other identities, that the path-matched income calculation reproduces the baseline ruin indicators exactly when each strategy is assigned its own 4% withdrawal. The bond-free strategies, the comparison at equal volatility, its sensitivity analyses ([eqvol_sensitivity.py](#)), resampled histories ([historical_panel_bootstrap_eqvol.py](#)) and managed-futures variants ([mf_variant_lifecycle.py](#)), the Sharpe ratios in excess of bills, and the comparison of the managed-futures series with commercial indexes by subperiod are produced by [a separate script](#), which reuses the same seed and reproduces the baseline results exactly. The [build instructions](#) describe how to regenerate the tables and figures from the result files.

References

- Anarkulova, Aizhan, Scott Cederburg, and Michael S. O’Doherty, 2025, Beyond the Status Quo: A Critical Assessment of Lifecycle Investment Advice, Working paper, July 10 revision.
- BarclayHedge, 2026, Barclay BTOP50 Index, <https://www.barclayhedge.com/solutions/indices/btop50-index/>.
- Board of Governors of the Federal Reserve System, 2026, Foreign Exchange Rates—H.10 Release, <https://www.federalreserve.gov/releases/h10/>.
- CME Group, 2024, Index Options Box Spreads as Financing Tool, <https://www.cmegroup.com/articles/2024/index-options-box-spreads-as-financing-tool.html>.
- 2026, Products: First Day of Trading, <https://www.cmegroup.com/media-room/historical-first-trade-dates.html>.
- Federal Reserve Bank of St. Louis, 2026, Federal Reserve Economic Data, <https://fred.stlouisfed.org/>.
- Financial Industry Regulatory Authority, 2021, Regulatory Notice 21-12: Margin, <https://www.finra.org/rules-guidance/notices/21-12>.
- Glasserman, Paul, 1991, Structural Conditions for Perturbation Analysis Derivative Estimation: Finite-Time Performance Indices, *Operations Research* 39, 724–738.
- Internal Revenue Service, 2026, Topic No. 409: Capital Gains and Losses, <https://www.irs.gov/taxtopics/tc409>.
- Jordà, Òscar, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick, and Alan M. Taylor, 2019, The Rate of Return on Everything, 1870–2015, *Quarterly Journal of Economics* 134, 1225–1298.
- 2023, The Jordà–Schularick–Taylor Macrohistory Database, Release 6, <https://www.macrohistory.net/database/>.
- Kuvshinov, Dmitry, and Kaspar Zimmermann, 2022, The Big Bang: Stock Market Capitalization in the Long Run, *Journal of Financial Economics* 145, 527–552.
- Litman Gregory Funds Trust, 2026, iMGP DBi Managed Futures Strategy ETF: Summary Prospectus, <https://www.sec.gov/Archives/edgar/data/1020425/000119312526197034/d62276d497k.htm>.
- London Bullion Market Association, 2025, LBMA Gold Price, afternoon (PM) fixing, <https://www.lbma.org.uk/prices-and-data/precious-metal-prices>.
- Maillard, Sébastien, Thierry Roncalli, and Jérôme Teiletche, 2010, The Properties of Equally Weighted Risk Contribution Portfolios, *Journal of Portfolio Management* 36, 60–70.
- Müller, Karsten, Chenzi Xu, Mohamed Lehibib, and Ziliang Chen, 2025, The Global Macro Database: A New International Macroeconomic Dataset, Working Paper, Used for 2021–2025 inflation and short rates; <https://www.globalmacrodata.com/>, National Bureau of Economic Research.
- National Bureau of Economic Research, 2016, NBER Macrohistory Database, Distributed via FRED, Federal Reserve Bank of St. Louis.
- Officer, Lawrence H., and Samuel H. Williamson, 2024, The Price of Gold, 1257–Present, and Exchange Rates Between the United States Dollar and Forty-one Currencies, *MeasuringWorth*.
- Organisation for Economic Co-operation and Development, 2026, Main Economic Indicators, Distributed through Federal Reserve Economic Data (FRED).
- Politis, Dimitris N., and Joseph P. Romano, 1994, The Stationary Bootstrap, *Journal of the American Statistical Association* 89, 1303–1313.
- Roncalli, Thierry, 2014, Introduction to Risk Parity and Budgeting, <https://arxiv.org/abs/1403.1889>.

Société Générale, 2026, SG CTA Index, SG Trend Index, and SG Trend Indicator, <https://wholesale.banking.societegenerale.com/en/prime-services-indices/>.

SPDR Gold Trust, 2022, 2022 Grantor Trust Tax Reporting Statement, <https://www.spdrgoldshares.com/media/GLD/file/SPDR-Gold-Trust-Tax-Information-2022.pdf>.

Testfolio, 2026, Simulated Tickers and Preset Return Series, <https://testfol.io/simulated-tickers>.

United States Mint, 1974, Statement on Gold Clause Resolution, <https://www.usmint.gov/learn/history/historical-documents/statement-on-gold-clause-resolution>.

Wang, Joseph, 2025, Long-Dated Box Spreads: A Better Way to Buy a Home, Cboe, Guest Author Series, <https://www.cboe.com/insights/posts/long-dated-box-spreads-a-better-way-to-buy-a-home-updated/>.

World Bank, 2025, Commodity Markets “Pink Sheet” Monthly Prices, <https://www.worldbank.org/en/research/commodity-markets>.

Yahoo Finance, 2026, Historical daily prices for listed managed-futures funds, <https://finance.yahoo.com>.